

# Homework 4

EE 999 — Introduction to Control Systems

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*Due Wednesday, October 7, 2026*

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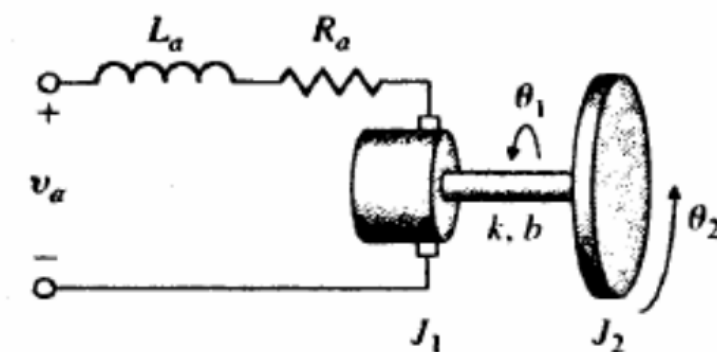
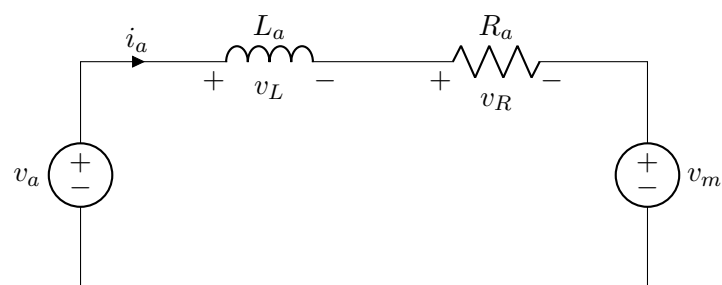
**Problem A.1** — Equations of Motion
Figure 1: DC motor driving a load inertia  $J_2$ 

Figure 2: Electrical Circuit Model

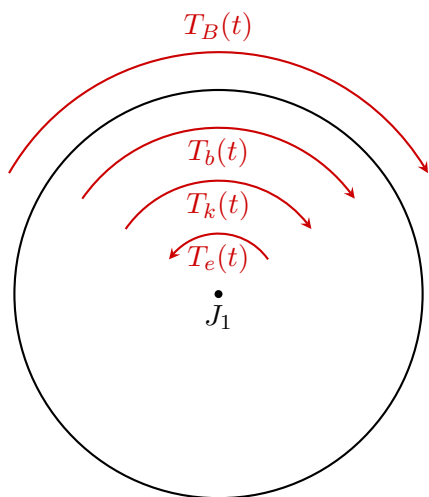


Figure 3: Free Body Diagram of Rotor

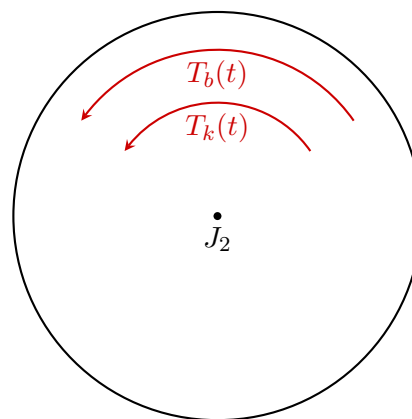


Figure 4: Free Body Diagram of Load

Electrical side:

$$v_a(t) = v_L(t) + v_R(t) + v_m(t)$$

$$v_L(t) = L_a \dot{i}_a(t)$$

$$v_R(t) = R_a i_a(t)$$

$$v_m(t) = K_e \dot{\theta}_1(t)$$

Rotor:

$$J_1 \ddot{\theta}_1(t) = T_{net.J_1}(t) = T_e(t) - T_B(t) - T_k(t) - T_b(t)$$

$$T_e(t) = K_t i_a(t)$$

$$T_B(t) = B \dot{\theta}_1(t)$$

$$T_k(t) = k(\theta_1 - \theta_2)$$

$$T_b(t) = b(\dot{\theta}_1 - \dot{\theta}_2)$$

Load:

$$J_2 \ddot{\theta}_2(t) = T_{net.J_2}(t) = T_b(t) + T_k(t)$$

Standard Forms

$$v_a(t) = L_a \dot{i}_a(t) + R_a i_a(t) + K_e \dot{\theta}_1(t)$$

$$J_1 \ddot{\theta}_1(t) = K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$$

$$J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + b \dot{\theta}_1 + k \theta_1(t) = K_t i_a(t) + k \theta_2 + b \dot{\theta}_2$$

$$J_2 \ddot{\theta}_2(t) = k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t))$$

$$J_2 \ddot{\theta}_2(t) + b \dot{\theta}_2(t) + k \theta_2(t) = k \theta_1(t) + b \dot{\theta}_1(t)$$

**Problem A.2** — Electrical Circuit Model

| Mechanical               | Units             | Electrical         | Units |
|--------------------------|-------------------|--------------------|-------|
| $\dot{\theta}(t) = v(t)$ | rad/s             | Voltage            | V     |
| $T(t) = i(t)$            | N·m               | Current            | A     |
| $J = C$                  | kg·m <sup>2</sup> | Capacitance        | F     |
| $B = 1/R$                | N·m·s/rad         | Conductance        | S     |
| $k = 1/L$                | N·m/rad           | Inverse inductance | 1/H   |

Table 1: Firestone analogy: signal and unit correspondence.

| Mechanical                             | Electrical                  |
|----------------------------------------|-----------------------------|
| $T = J\ddot{\theta}$                   | $i = C\dot{v}$              |
| $T = B\dot{\theta}$                    | $i = v/R$                   |
| $T = k\theta = k \int \dot{\theta} dt$ | $i = \frac{1}{L} \int v dt$ |

Table 2: Element laws under the Firestone analogy.

Current equation for the first inertia:

$$\begin{aligned}
 J_1 \ddot{\theta}_1(t) &= K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2) \\
 k_t i_a(t) &= J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + k(\theta_1 - \theta_2) + b(\dot{\theta}_1 - \dot{\theta}_2) \\
 I_e(t) &= C_1 \dot{v}_1(t) + \frac{1}{R_B} v_1(t) + \frac{1}{L_k} \left( \int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

and for the second:

$$\begin{aligned}
 J_2 \ddot{\theta}_2(t) &= k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t)) \\
 J_2 \ddot{\theta}_2(t) &= \frac{1}{L_k} \left( \int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

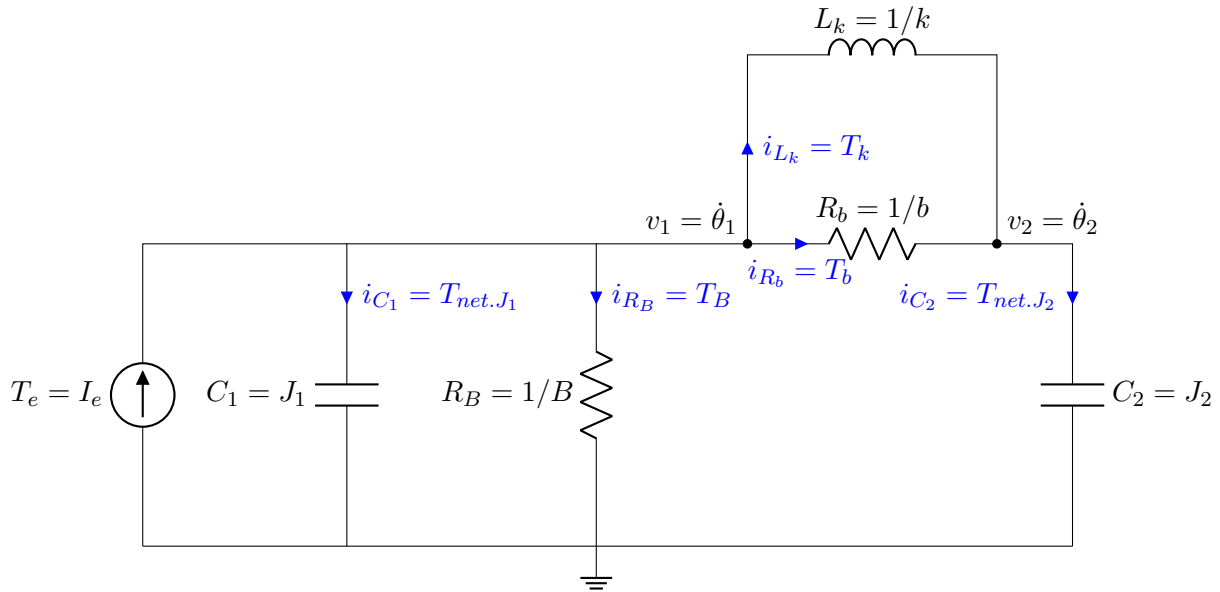


Figure 5: Firestone analogous circuit, with branch currents

**Problem B — Units Balancing**

$$J_m \ddot{\theta}_m + b \dot{\theta}_m = K_t i_a$$

$$(kg \cdot m^2) \left( \frac{rad}{s^2} \right) + \left( \frac{N \cdot m \cdot s}{rad} \right) \left( \frac{rad}{s} \right) = \left( \frac{N \cdot m}{A} \right) (A)$$

$$(N \cdot m \cdot \cancel{s}) \left( \frac{rad}{\cancel{s} s^2} \right) + \left( \frac{N \cdot m \cdot \cancel{s}}{rad} \right) \left( \frac{rad}{\cancel{s} s} \right) = \left( \frac{N \cdot m}{A} \right) (\cancel{A}) = N \cdot m$$

Noting that  $K_t$  is the motor torque constant with units of  $\frac{N \cdot m}{A}$ , so the amps cancel, and radians are eliminated as being unitless.

**Problem C.2** — Linearization

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Equation given in problem:

$$q(t) = 20 \frac{e^{x(t)/15} - 1}{e^{x(t)/15} + 1} \quad (1)$$

Linearization equations:

$$q(t) = f(x(t)) \quad (2)$$

$$x(t) = x_0 + \delta x(t) \quad (3)$$

$$q(t) = q_0 + \delta q(t) \quad (4)$$

$$\delta q(t) \simeq K_{xq} \delta x(t) \quad (5)$$

$$q(t) \simeq q_0 + K_{xq} \delta x(t) \quad (6)$$

To find  $\delta x(t)$ , take the derivative

$$\begin{aligned} u(t) &= e^{x(t)/15} - 1 \\ \dot{u}(t) &= \frac{1}{15} e^{x(t)/15} \end{aligned} \quad (7)$$

$$\begin{aligned} v(t) &= e^{x(t)/15} + 1 \\ \dot{v}(t) &= \frac{1}{15} e^{x(t)/15} \end{aligned} \quad (8)$$

$$q(t) = 20 \frac{u(t)}{v(t)} \quad (9)$$

$$\begin{aligned} \dot{q}(t) &= 20 \frac{v\dot{u} - u\dot{v}}{v^2} \\ \dot{q}(t) &= 20 \frac{\frac{1}{15} e^{x/15} (e^{x/15} + 1) - \frac{1}{15} e^{x/15} (e^{x/15} - 1)}{(e^{x/15} + 1)^2} \\ \dot{q}(t) &= \frac{4}{3} \frac{2e^{x/15}}{(e^{x/15} + 1)^2} \end{aligned}$$